

Problem C. Negative Strips

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

Given n, m and k , construct an $n \times m$ integer matrix such that:

- the sum of all elements of the matrix is **positive**;
- the sum of every k consecutive elements in the same row is **negative**;
- the sum of every k consecutive elements in the same column is **negative**.

If $k > m$, a row has no k consecutive elements; if $k > n$, a column has no k consecutive elements.

Input

A single line containing three integers n, m and k ($1 \leq n, m, k \leq 10^6, n \cdot m \leq 10^6$).

Output

If no such matrix exists, print No.

Otherwise, print **Yes** on the first line, followed by n lines with m integers each: the matrix. The absolute value of every element must be at most 10^9 . If there are multiple answers, print any of them.

Examples

standard input	standard output
3 5 4	Yes 1 -1 4 -5 1 4 -1 -1 -4 5 1 -4 -1 -1 4
4 5 3	Yes 2 8 -11 -1 11 -5 -3 7 -5 -3 -1 -6 3 2 -9 5 8 -14 2 11
4 6 3	No

Note

In the first example, every 4 consecutive elements in a row have a negative sum, the sum of all elements is 2, and the condition on columns is empty because $k > n$.

In the second example, every 3 consecutive elements in a row or in a column have a negative sum, and the sum of all elements is 1.